

Implementation and Testing of the Database, Application and Communication Layers

Objective

The objective of this practical was to implement and test the provided application by linking the database, application, and communication layers and verifying that all components communicate successfully and exchange data correctly.

Software Used:

- XAMPP Control Panel
- Apache Web Server
- MySQL Database
- phpMyAdmin
- Visual Studio Code
- PHP
- Google Chrome (Web Browser)

Implementation

Step 1: Database Layer

A MySQL database named phpcrud1 was created using phpMyAdmin. A table named tblusers was created with the following fields:

- ID (Primary Key, Auto Increment)
- FirstName
- LastName
- MobileNumber
- Email
- Address
- CreationDate

The database provides permanent storage for all user records.

Step 2: Application Layer

The PHP application was developed using Visual Studio Code. The following files were created:

dbconnection.php

index.php

insert.php

edit.php

read.php

The application connects to the database using the MySQLi extension through the dbconnection.php file.

Step 3: Communication Layer

Communication between the application and database was established through PHP and MySQL.

The connection was configured as follows:

Host: localhost

Username: root

Password: (blank)

Database: phpcrud1

Apache handled HTTP requests while MySQL processed all database queries.

Testing

The system was tested after implementation.

Test 1: Database Connection

The application successfully connected to the MySQL database without connection errors.

Status: Successful

Test 2: Insert Data

A new user record was entered through the web form.

The information was successfully saved into the tblusers table.

Status: Successful

Test 3: Read Data

The application successfully retrieved all records from the database and displayed them on the home page.

Status: Successful

Test 4: Update Data

An existing record was edited and the updated information was successfully saved.

Status: Successful

Test 5: Delete Data

A selected record was deleted successfully from the database.

Status: Successful

Verification of Communication

Communication between all system components was verified successfully.

The web browser communicated with the PHP application through Apache.

The PHP application communicated with the MySQL database using MySQLi.

Data entered by the user was stored correctly in the database.

Stored records were retrieved and displayed correctly.

Updated information was reflected immediately.

Deleted records were removed successfully.

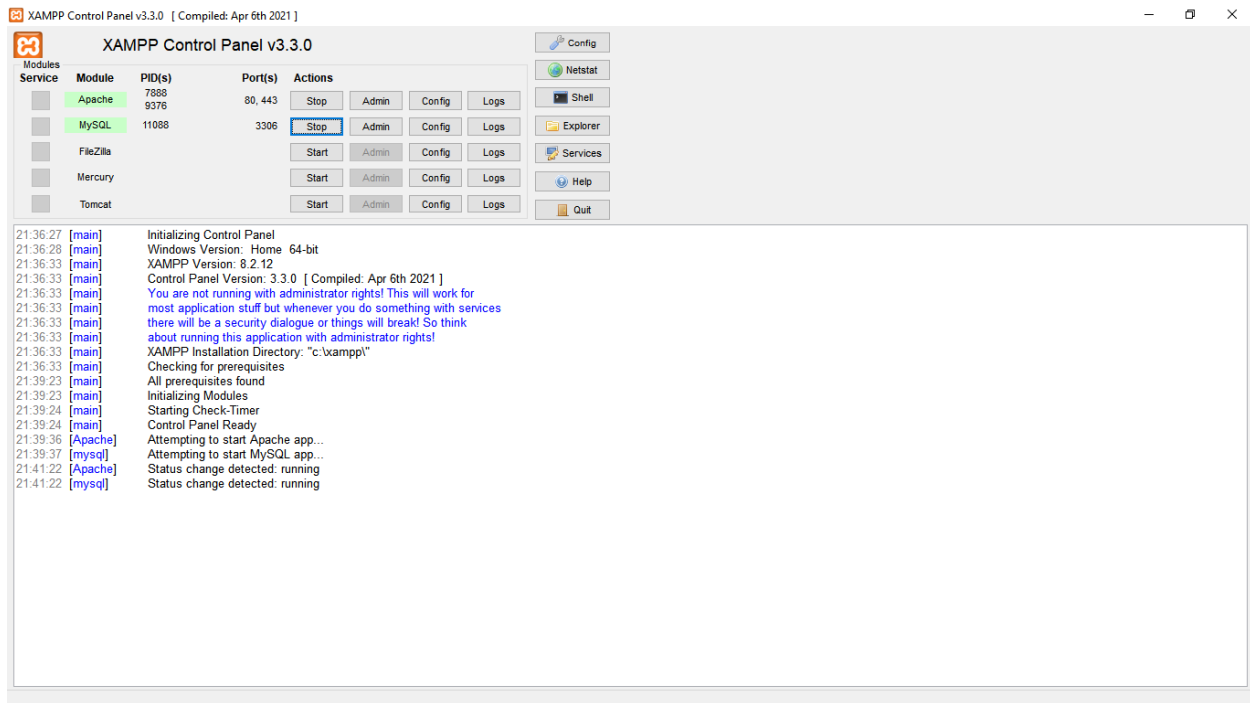
These results confirmed successful communication and data exchange between the database, application, and communication layers.

Conclusion

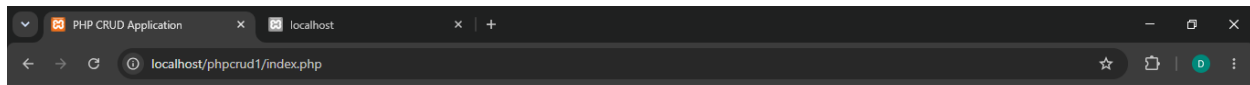
The implementation was completed successfully. The database, application, and communication layers were integrated correctly, allowing the application to perform Create, Read, Update, and Delete (CRUD) operations. Testing confirmed that all components communicated successfully and exchanged data accurately, demonstrating a functional distributed web application running on the localhost environment.

SCREENSHOTS OF:

XAMMP



THE CRUD APPLICATION HOME PAGE



User Management System

[Add New User](#)

ID	First Name	Last Name	Mobile	Email	Address	Creation Date	Actions
1	DERICK	Kemboi	779908007	kemboiderick3@gmail.com	NYERI	2026-06-29 21:19:59	View Edit Delete
2	TIMO	KIPLAGAT	789898989	KIPLAGATKKK7@gmail.com	NAKURU	2026-06-29 21:46:46	View Edit Delete

INSERTING A RECORD

PHP Crud Operation!!

localhost / 127.0.0.1 / phpcrud1

(31) WhatsApp

localhost/phpcrud1/insert.php

Fill Data

Fill below form.

First Name

Last Name

Enter your Mobile Number

Enter your Email id

Enter Your Address

Submit

View Already Inserted Data!! [View](#)

VIEWING RECORDS

Crud Operation Using PHP and

localhost

localhost/phpcrud1/read.php?viewid=2

User Details

First Name	TIMO	Last Name	KIPLAGAT
Email	KIPLAGATKKK7@gmail.com	Mobile Number	789898989
Address	NAKURU	Creation Date	2026-06-29 21:46:46

EDITING A RECORD

PHP Crud Operation!!

localhost

localhost/phpcrud1/edit.php?editid=2

Update

Update your info.

TIMO

MWANGI

789898989

KIPLAGATKKK7@gmail.com

KERICHO

Update

DELETING A RECORD-DELETED DERICK

PHP CRUD Application

localhost

localhost/phpcrud1/index.php

User Management System

Add New User

ID	First Name	Last Name	Mobile	Email	Address	Creation Date	Actions
2	TIMO	MWANGI	789898989	KIPLAGATKKK7@gmail.com	KERICHO	2026-06-29 21:49:12	ViewEditDelete

PHPMYADMIN HOING TBLUSERS

PHP CRUD Applicationlocalhost / 127.0.0.1 / phpcrud1WhatsApp

localhost/phpmyadmin/index.php?route=/sql&db=phpcrud1&table=tblusers&pos=0

Server: 127.0.0.1 » Database: phpcrud1 » Table: tblusers

BrowseStructureSQLSearchInsertExportImportPrivilegesOperationsTrackingTriggers

Showing rows 0 - 0 (1 total, Query took 0.0305 seconds)

SELECT * FROM `tblusers`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

IDFirstNameLastNameMobileNumberEmailAddressCreationDate

☐

Edit

Copy

Delete

2TIMOMWANGI789898989KIPLAGATKKK7@gmail.comKERICHO2026-06-29 21:49:12

Check all

With selected:

Edit

Copy

Delete

Export

Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

Print

Copy to clipboard

Export

Display chart

Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Bookmark this SQL query

Console